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Cyclopropenimines for activation of carbon dioxide

Abstract

A process, apparatus, and material for generating polymers are disclosed. Generating the polymers comprises reacting carbon dioxide (CO.sub.2) with a cyclopropenimine (CPI). Generating the polymers further comprises reacting monomers with a product of the reaction between the CPI and the CO.sub.2.

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Background/Summary

(1) The following disclosure is submitted under 35 U.S.C. 102(b)(1)(A):
(2) DISCLOSURE: Wu, D., Park, N., Hedrick, J., Campos, L. “Universal Reagents for CO_2 Capture, Storage, and Upcycling into Value-Added Polymer” [Conference Presentation Abstract] 264.sup.th American Chemical Society National Meeting and Exposition, Chicago, IL and Online, Abstract accepted for publication May 26, 2022.
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BACKGROUND
(4) The present disclosure relates to carbon dioxide (CO_2) capture and transfer and, more specifically, to generating polymers using CO_2 activated by cyclopropenimine (CPI) compounds.
(5) Techniques for capturing atmospheric CO_2 (e.g., direct-air-capture (DAC)) can be used to offset CO_2 emissions. Current DAC technologies generally involve sorption materials, which can absorb CO_2 gas at atmospheric levels and then desorb the gas as an isolated stream in specified intervals. Techniques for transferring and chemically transforming CO_2 can be used to produce synthetically useful compounds. For example, captured CO_2 may be used as a feedstock in the synthesis of polymeric materials. Upcycling CO_2 into useful monomers may also facilitate a shift in production away from standard, fossil fuel intensive approaches that employ highly toxic chemicals, such as phosgene.
SUMMARY
(6) Various embodiments are directed to a process of generating polymers. Generating the polymers comprises reacting carbon dioxide (CO_2) with a cyclopropenimine (CPI) and reacting monomers with a product of the reacting. This can provide a way to capture CO_2 and generate useful chemicals from renewable and abundant sources. The CO_2 may be obtained from air. This may be advantageous because air can be inexpensive and easily obtained. In some embodiments, the process generates polycarbonates or polyurethanes. There is a high demand for these polymers. Reacting the CO_2 with the CPI can include adding the CO_2 to a suspension of the CPI in a polar organic solvent. This may allow the reaction to be carried out under mild conditions with readily available solvents.
(7) Further embodiments are directed to an apparatus and a CPI material for carrying out the method of generating metal carbonates.
(8) The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawings included in the present application are incorporated into, and form part of, the Specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.
(2) FIG. **1** is a chemical reaction diagram illustrating a process of forming a cyclopropenimine (CPI), according to some embodiments.
(3) FIG. **2** is a chemical reaction diagram illustrating a process of forming a CPI-carbon dioxide (CPI-CO_2) adduct, according to some embodiments.
(4) FIG. **3** illustrates experimental examples of a process of CPI formation, according to some embodiments.
(5) FIG. **4**A is a chemical structure diagram illustrating CPIs with a series of R groups, according to some embodiments.
(6) FIG. **4**B is a chemical structure diagram illustrating CPIs with a series of R′ groups, according to some embodiments.
(7) FIG. **5** is a chemical reaction diagram illustrating a process of forming a CPI-functionalized polymethacrylate, according to some embodiments.
(8) FIG. **6** is a chemical reaction diagram illustrating processes of forming a CPI-functionalized polystyrene, according to some embodiments.
(9) FIG. **7** is a chemical reaction diagram illustrating a process of forming a CPI-functionalized polyurethane, according to some embodiments.
(10) FIGS. **8**A and **8**B are a chemical reaction diagrams illustrating processes of CPI-facilitated mineralization,

according to some embodiments.
(11) FIGS. **9**A and **9**B are chemical reaction diagrams illustrating generic processes of CPI-facilitated polymerization, according to some embodiments.
(12) FIG. **9**C is a chemical reaction diagram illustrating a process of CPI recovery, according to some embodiments.
(13) FIG. **10** is a chemical reaction diagram illustrating example processes of CPI-facilitated polymerization, according to some embodiments.
(14) FIGS. **11**A and **11**B are tables illustrating chemical structures corresponding to experimental examples of CPI-facilitated polymerizations, according to some embodiments.
(15) FIG. **12** is a table illustrating experimental data characterizing polymer products formed in the experimental examples of FIGS. **11**A and **11**B.
(16) FIG. **13** is a table illustrating chemical structures corresponding to an experimental example of CPI-facilitated polymerization, according to some embodiments.
(17) FIG. **14** is a table illustrating experimental data characterizing a polymer illustrated in FIG. **13**.
(18) While the invention is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings, and will be described in detail. It should be understood, however, that the intention is not to limit the invention to the particular embodiments described. Instead, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention.

DETAILED DESCRIPTION

(19) Embodiments of the present invention are generally directed to carbon dioxide (CO.sub.2) capture and transfer and, more specifically, to generating polymers using CO.sub.2 activated by cyclopropenimine (CPI) compounds. While the present disclosure is not necessarily limited to such applications, various aspects of the disclosure may be appreciated through a discussion of examples using this context.
(20) Techniques for reducing atmospheric CO.sub.2 are essential for the goal of limiting the global temperature rise to 1.5° C. by 2050. Current emissions at about 35 metric gigatons per year (Gt/yr) are expected to rise to ˜40-45 Gt/yr by 2050. Point source capture, zero-emission technologies, such as renewables for energy production, and reduced-emission programs are expected to lower emissions (e.g., by about 800-900 Mt/yr). However, these efforts cannot offset CO.sub.2 from long distance travel/cargo transport and certain heavy industries (expected to account for 15+% emissions annually), nor can they remove already-emitted CO.sub.2 from the atmosphere.
(21) Negative emissions using DAC may overcome these challenges. Current DAC technologies generally involve sorption materials, which can absorb CO.sub.2 gas at atmospheric levels and then desorb the gas as an isolated stream in specified intervals. Upcycling CO.sub.2 into useful monomers would also facilitate a shift in production away from standard, fossil fuel intensive approaches that employ highly toxic chemicals, such as phosgene. However, challenges remain in scaling DAC sufficiently. For example, current atmospheric loading of CO.sub.2 is a dilute 415 ppm, but the estimated total carbon load in the atmosphere is 900+ Gt.
(22) Current chemisorption technologies used for point-of-generation capture of about 5-30% by volume can require 2-4 gigajoules (GJ) per metric ton (t) of CO.sub.2, while existing commercially available DAC technologies can use ˜12 GJ/t of CO.sub.2 at atmospheric concentrations. The regeneration energy needed to release CO.sub.2 from capture reagents dominates the energy costs. As an example, current amine-scrubbing plants use aqueous solutions that capture CO.sub.2 at about 25-40° C. and release CO.sub.2 at about 100-150° C., with flow rates of thousands of tons per hour, thus requiring heating and cooling of significant quantities of fluid. Recent approaches to CO.sub.2 capture, including solid-phase chemisorption and water-lean systems, have not sufficiently reduced these regeneration energies and can suffer from scalability and stability issues. A key chemical challenge remains to produce new molecules that can capture CO.sub.2 and regenerate/release the captured CO.sub.2 with a minimum energy budget.
(23) Taking the challenge beyond lowering the energetic penalties in DAC, the ability to convert CO.sub.2 into useful polymers, minerals, functional molecules, biodegradable plastics, and consumable products at scale demands the development of new chemistries that allow efficient CO.sub.2 capture and require less energy for release/storage. Therefore, we face an urgent need for new catalysts and reagents that operate efficiently at or about standard/room temperature and pressure, while reducing the overall regeneration energy costs. Such molecules could enable DAC systems that operate under dilute concentrations of CO.sub.2 (e.g., ambient conditions).
(24) There is also a need for new ways of efficiently generating common polymers (e.g., polyurethanes, polycarbonates, etc.) without requiring hazardous or non-renewable materials. For example, polyurethanes (PUs) are considered highly versatile and one of the most important classes of polymeric materials. Unfortunately, conventional polyurethane synthesis necessitates the use of isocyanate monomers, which can be toxic and sensitive to moisture. Because of these and other disadvantages, there is an emerging interest in finding a sustainable approach toward non-isocyanate polyurethanes (NIPUs).
(25) Various embodiments of the present disclosure may be used to overcome these and other challenges. Disclosed herein are cyclopropenimine (CPI)-based molecules and polymers ("CPIs") that may be used for upcycling and capture of CO.sub.2. For example, CPIs can act as CO.sub.2-transfer/activation agents for permanent storage of

CO.sub.2 in a process of mineralization from sea water. CPIs can also be used to activate CO.sub.2 to be used in the synthesis of polymers such as polycarbonates and polyurethanes without isocyanate. In some embodiments, the CPIs can capture CO.sub.2 from dilute sources under mild conditions without requiring an external stimulus (e.g., heat, electrochemical stimuli, etc.). This CO.sub.2 capture results in formation of adducts with CO.sub.2 (CPI-CO.sub.2 adducts), which can "activate" CO.sub.2 for subsequent chemical transformations. In some embodiments, the activated CO.sub.2 (e.g., carbonate, bicarbonate, etc.) can be released from the CPI-CO.sub.2 adducts for regeneration/restoration of the CPIs in a low energy semi-automated process.
(26) The disclosed CPIs can be superbasic (e.g., having conjugate acids with pK.sub.BH+-27-28) due to aromatic stabilization from the cyclopropenium ion. Additionally, the CPIs may be synthesized on a large scale from readily available, inexpensive substrates and processed into emulsions, membranes, particles, etc. that may be integrated within CO.sub.2 reactors. Such materials may enable DAC systems that operate at ambient conditions using localized, renewable energy sources.
(27) Referring now to the drawings, in which like numerals represent the same or similar elements, FIG. **1** is a chemical reaction diagram illustrating a process **100** of forming a CPI **110**, according to some embodiments. Herein, "CPI" refers to a bis(amino) CPI species unless otherwise indicated. The CPI **110** illustrated in FIG. **1** can be a CPI molecule where R and R′ are organic substituents (see below). Process **100** can involve a first step **103** wherein tetrachlorocyclopropene is reacted with a secondary amine R.sub.2NH (e.g., dicyclohexylamine, where R=cyclohexyl ("Cy")) in a solvent such as dichloromethane (DCM). In a second step **106**, the resulting diamine chloride salt can be reacted with a primary amine R′—NH.sub.2 to form a CPI conjugate acid chloride (CPI-H.sup.+/Cl.sup.−). In a third step **109**, basification of the conjugate acid CPI-H.sup.+ yields the CPI. The ability to vary R and R′ groups based on secondary and primary amine selection provides modularity to the CPI core by allowing tuning of parameters such as reactivity, solubility, etc. Examples and effects of R and R′ groups are discussed in greater detail below.
(28) FIG. **2** is a chemical reaction diagram illustrating a process **200** of forming a CPI-CO.sub.2 adduct **120**, according to some embodiments. In process **200**, CPI **110** can be reacted with CO.sub.2 gas to form the CPI-CO.sub.2 adduct **120**. The CO.sub.2 can come from direct sources such as substantially pure CO.sub.2 (e.g., ˜100% CO.sub.2), CO.sub.2/inert gas mixtures (e.g., ˜20-99% CO.sub.2), etc. or dilute sources such as atmospheric gas (e.g., ˜0.04% CO.sub.2), exhaled air (e.g., ˜4% CO.sub.2), etc. This reaction may be carried out under ambient conditions. The CPI-CO.sub.2 adduct **120** is a zwitterion characterized by an aromatic cyclopropenium (tris(amino)cyclopropenium, or TAC) ion, which can be stabilized by electron-donating amine substituents, and an anionic carbonate. The activated CO.sub.2 may be released in a "CO.sub.2 transfer" reaction resulting in a carbonate anion (CO.sub.3.sup.−) or carbonate ester product (e.g., polycarbonate, polyurethane, etc.) and a conjugate acid of the CPI (CPI-H.sup.+ **130**). Reactants involved in the CO.sub.2 transfer operation are not shown in FIG. **2**. However, examples of CO.sub.2 transfer reactions are illustrated in FIGS. **8**A-**12**B).
(29) Upon loss of CO.sub.2.sup.−, CPI **110** may be recovered by basification of CPI-H.sup.+ **130**. In some embodiments, CPI recovery can include a biphasic process discussed in greater detail below (see, e.g., FIG. **8**B). The recovered CPI **110** can be reused to react with additional CO.sub.2. This is discussed in greater detail with respect to FIGS. **8**A-**9**B. The aromatic stabilization and synthetic modularity of CPIs have enabled their use in catalytic transformations, including Michael additions, Mannich reactions, Wittig rearrangements, and ring-opening polymerization. The cyclopropenium conjugate acids of CPIs have also been utilized as Brønsted acid catalysts in reactions involving additions to oxocarbeniums and hydroamination of alkenes.
(30) FIG. **3** illustrates experimental examples of a process **300** of CPI formation, according to some embodiments. In this example, a diamine was mixed with a selected primary amine (R′—NH.sub.2) and N,N′-diisopropylethylamine (DIPEA) in dichloromethane (CH.sub.2Cl.sub.2) and allowed to react overnight (e.g., step **106** of process **100**). The reaction mixture was then washed with 1 M HCl, followed by basification with 1 M NaOH to convert the conjugate acid CPI-H.sup.+ (not shown) into the CPI (e.g., step **109** of process **100**). Process **300** was repeated using different R′—NH.sub.2 reactants under substantially similar reaction conditions in order to form a series of CPIs **320**, which are shown with their corresponding percent yields. Herein, "Me" represents a methyl group, and "Ph" represents a phenyl group.
(31) FIG. **4**A is a chemical structure diagram illustrating CPIs **400** with a series of R groups, according to some embodiments. The R groups can be modified to tune solubility, basicity, stability, etc. For example, CPI **410**, which has morpholino R groups, may be more soluble than CPI **420** and CPI **430**, which respectively have isopropyl and cyclohexyl R groups. Further, the R groups may be modified to induce twisting of the plane between the cyclopropene ring and the —NR.sub.2 moieties due to steric interactions. This may impact the basicity and reactivity of the CPIs. The R′ group of CPIs **410-430** can be any appropriate organic substituent (see, e.g., FIGS. **3** and **4**B).
(32) FIG. **4**B is a chemical structure diagram illustrating CPIs **401** with a series of R′ groups, according to some embodiments. The R′ groups of CPI **440** and CPI **450**, respectively, include linear alkyl and cyclic aromatic moieties. The R′ group of CPI **460** includes an alcohol moiety. In CPIs **440-460**, R.sup.1 and R.sup.2 can, independently, be

hydrogen atoms or any appropriate reactive or unreactive functional groups. The R′ groups can be modified to tune interactions with CO_2. For example, when R′ includes an alcohol moiety (e.g., CPI **460** or CPIs **440**/**450** when R^1 has hydroxyl group), a synergistic effect on CO_2 capture analogous to alcohol-containing guanidine and amidine systems may be conferred. The R groups of CPIs **440**-**470** can be any appropriate organic substituents (see, e.g., FIG. **4**A).
(33) Further embodiments can include polymers containing CPI and/or tris[amino]cyclopropenium (TAC) pendant groups. These polymers can be formed using polymerizable building blocks containing pendant CPIs and TAC ions. In other embodiments, CPI and/or TAC pendant groups may be added post-polymerization. A wide variety of polymers may be synthesized with these pendant groups, such as polynorbornenes, polycarbonates, polystyrenes, polymethylmethacrylates, polymethacrylates, polyethers, polyesters, epoxide resins, polyamines, etc.
(34) In some embodiments, CPI-functionalized methacrylate polymers can be formed by reacting amine functionalized methacrylates or methacrylamides with various chloro-CPI precursors to generate an array of methacrylate CPI salts. For example, TAC monomers can be formed using a primary amine R′—NH_2 (e.g., primary amine at operation **106** of process **100**) where R′ includes a methacrylate moiety. TAC monomers may undergo polymerization and copolymerization by a variety of reversible deactivation radical polymerization (RDRP) reactions.
(35) FIG. **5** is a chemical reaction diagram illustrating a process **500** of forming a CPI-functionalized polymethacrylate, according to some embodiments. In this example, a TAC monomer **510** is provided. The TAC monomer has a TAC ion moiety and can be formed, for example, in process **100** using 2-aminoethylmethacrylamide. The TAC monomer **510** can be polymerized (e.g., via RDRP) and, following polymerization, neutralized to the free-base CPI. This yields a CPI polymer **520**. While FIG. **5** illustrates a homopolymer **520**, the TAC monomer **510** can be copolymerized with other monomers, such as methacrylates having various functional groups. This can allow CPI reactivity to be tuned by utilizing the effects of neighboring pendant groups (e.g., hydroxyl groups).
(36) FIG. **6** is a chemical reaction diagram illustrating processes **601** and **602** of forming CPI-functionalized polystyrenes, according to some embodiments. Processes **601** and **602** begin with styrene-TAC monomers **610** and **620**, respectively. Polymerization of these monomers affords corresponding cationic TAC-functionalized polystyrenes (not shown), which can be neutralized by basification to yield CPI-functionalized polystyrenes **630** (process **601**) and **640** (process **602**). In process **602**, deprotection of the TAC t-butyl ester results in a zwitterionic polystyrene **640**.
(37) In further embodiments, a series of CPI-functionalized styrene monomers may be produced and used to generate solid phase resins that may be incorporated into continuous capture and transformation processes.
(38) FIG. **7** is a chemical reaction diagram illustrating a process **700** of forming CPI-functionalized polyurethanes, according to some embodiments. In some embodiments, amino-alcohols can be used to form TAC-diol monomers **710** for organocatalyzed polyadditions, providing CPI-functionalized polyurethanes **720** following neutralization. For example, serinol can be reacted with a chloro-CPI to form the TAC-diol monomer **710** illustrated in FIG. **7**. Polyaddition with a diisocyanate can afford a corresponding TAC-functionalized polyurethane (not shown). In structure **720**, the diisocyanate R′ can be any appropriate organic moiety selected independently of the CPI organic groups. Neutralization by addition of a base can result in the illustrated CPI-functionalized polyurethane **720**. Polymerizations of TAC-diol monomers can be accomplished as homogeneous solutions, as emulsions or suspensions, or in bulk. In some embodiments, crosslinking agents can be used to form various CPI-functionalized polyurethanes. Using techniques such as these, TAC-diols can be used as building blocks for polymer precipitates, crosslinked polymer beads, or high surface area polymeric foams using conventional synthetic methodologies.
(39) FIGS. **8**A and **8**B are a chemical reaction diagrams illustrating processes **800** and **801** of CPI-facilitated mineralization, according to some embodiments. Processes **800** and **801** may be used to utilize and/or store captured CO_2. For example, large scale processes may be used for DAC, CO_2 storage, and/or conversion of CO_2 into more useful chemical species. In FIGS. **8**A and **8**B, CPI **810** is illustrated with a starred bond, which can be a covalent bond, e.g., to a carbon atom of R′ (see, e.g., FIGS. **3**-**4**B), a metal complex, a polymer/oligomer repeat unit (see, e.g., FIGS. **5**-**7**), etc. In some embodiments, at least one R group may be replaced by a bond to a metal complex or polymer repeat unit as well.
(40) Process **800** (FIG. **8**A) includes mineralization of CO_2. Various examples of CPIs **810** can be used in mineralization reactions such as process **800**, including small molecules, oligomers, polymers, etc. In these mineralization processes, CPI and CO_2 act as a carbonate source that can react with metal cations (e.g., from salts dissolved in water) to form carbonate minerals (or “metal carbonates”). In each of processes CPI **810** is reacted with gaseous CO_2 to form a CPI-CO_2 adduct **820**. This is illustrated at operation **803**. The CO_2 can come from direct sources such as substantially pure CO_2 (e.g., ˜100% CO_2), CO_2/inert gas mixtures (e.g., ˜20-99% CO_2), etc. or dilute sources such as atmospheric gas (e.g., ˜0.04% CO_2), exhaled air (e.g., ˜4% CO_2), etc. In the illustrated examples, CPI **810** can be suspended in a polar organic solvent such as acetonitrile. However, CPI **810** may be suspended in different solvents/liquids, polymer networks, solid supports, etc. In further embodiments, CPI **810** can be provided in a solvent-free environment. This is discussed in greater

detail below.
(41) At operation **806**, the adduct **820** can be mixed with an aqueous solution containing mono- and/or divalent metal cations (M^{n+}, where n indicates the oxidation state of the metal M). The cations can be provided as a salt MX or MX_2, such a metal halide (e.g., where X^- is chloride and M^{n+} is Na^+, K^+, Mg^{2+}, Ca^{2+}, or Sr^{2+}). This results in formation of a CPI conjugate acid halide salt CPI-H^+/X^- and a bicarbonate $MHCO_3$ when the metal is monovalent or a carbonate MCO_3 when the metal is divalent. The solid metal carbonate (or bicarbonate) can be removed from the mixture by filtration. At operation **809**, CPI-H^+ can be converted back to the free-base CPI **810** by basification (e.g., addition of an aqueous alkaline solution). The recovered CPI **810** can be isolated as a solid and/or reacted with additional CO_2 (returning to operation **803**). Examples of techniques for CPI recovery are discussed in greater detail below.
(42) In some embodiments, operation **806** can include mixing the CPI-CO_2 adduct **820** with brine (aqueous NaCl) in a “Solvay” process. This can result in formation of soluble CPI-H^+/Cl^- (**830**, where X^-=Cl^-) and solid sodium bicarbonate ($NaHCO_3$). The sodium bicarbonate can then be filtered and dried, and the CPI **810** can be recovered from the remaining solution by addition of a base (e.g., an ammonium hydroxide or sodium hydroxide solution) to the CH_3CN solution at operation **809**.
(43) An experimental example according to process **800** was carried out with a CPI (e.g., CPI **810**, where R=cyclohexyl and the starred bond is to n-butyl) suspended in CH_3CN. Gaseous CO_2 was added to the CPI suspension from a dilute source, exhaled air (human breath), which contains ˜4% CO_2 and ˜6% H_2O (operation **803**). The resulting CH_3CN-soluble species was mixed with brine (aqueous NaCl), resulting in a mixture of CPI-H^+/Cl^- and $NaHCO_3$ (operation **806**). The $NaHCO_3$ was filtered and dried, and the CPI-H^+/Cl^- was worked-up with ammonium hydroxide to provide the free-base CPI (operation **809**). This recovered CPI was isolated as a solid.
(44) FIG. **8**B is a chemical reaction diagram illustrating a process **801** of CPI-facilitated CO_2 mineralization and biphasic CPI recovery, according to some embodiments. At operation **813**, CPI **810** can be reacted with CO_2 using substantially the same techniques as at operation **803** (FIG. **8**A). The reaction with CO_2 forms a soluble CPI-H^+/HCO_3^- salt. Subsequent addition of aqueous MX_2 (e.g., $MgCl_2$, $CaCl_2$, or $SrCl_2$) leads to formation of a metal carbonate MCO_3 (e.g., $MgCO_3$, $CaCO_3$, or $SrCO_3$) and CPI-H^+/X^- salt at operation **816**. The reaction at operation **816** forms an organic phase **910** (soluble CPI-H^+/X^- salt **830**) and an aqueous phase **920** (insoluble MCO_3). In some embodiments, other examples of metal halide salts (MX or MX_2) may be used in process **801**. For example, $NaCl_{(aq)}$ can be reacted to form an aqueous phase **920** with $NaHCO_3$/Na_2CO_3 (M_2CO_3) under substantially the same conditions.
(45) The MCO_3 (or M_2CO_3) precipitate can be removed via filtration and drying of the aqueous phase **920**. CPI **810** can be substantially recovered from the organic phase **910** at operation **819**. In some embodiments, addition of an aqueous base such as NaOH or $NaNH_4$ (e.g., about 1 M) to the CPI-H^+/X^- salt solution **910** can regenerate CPI **810**. However, in other embodiments operation **819** may be carried out under milder conditions using a biphasic process. In these instances, the CPI-H^+/X^- salt solution **910** can undergo solvent exchange from the polar organic solvent (CH_3CN) into a non-polar solvent (e.g., toluene, hexanes, combinations thereof, etc.). Addition of an aqueous alkaline solution, such as about 1 mole/liter (M) Na_2CO_3(aq) or about 20-30 vol % $NH_4OH_{(aq)}$ (e.g., about 25%) can then regenerate the CPI moieties (CPI **810**). Experimental examples of process **801**, using either dilute or direct CO_2 sources, yielded at least 90-95% MCO_3 and at least 90-95% recovered CPI **810**. The recovered CPI can be used for additional CO_2 capture and activation. In some embodiments, CPI recovery processes such as those discussed above can be carried out after one or more CO_2 capture/activation cycles, thereby allowing reuse of CPI over multiple cycles.
(46) In some embodiments, the CPI-facilitated CO_2 capture/activation processes use seawater as the source of MX_2 at operation **806** (FIG. **8**A) or **816** (FIG. **8**B). Seawater is a mixture of about 96.5% water, 2.5% salts, and small amounts of other substances such as dissolved inorganic and organic materials, particulates, and atmospheric gases. The salts found in seawater can include salts of metal ions M^{n+} such as Mg^{2+}, Ca^{2+}, Sr^{2+}, and Na^+ in varying concentrations. The relative concentrations of different M^{n+} ions, pH, and other features of seawater can differ by region. Some of these variations may affect the mineralization products, as will be understood by those of ordinary skill in the art. In experimental examples of process **801** using seawater collected from the Atlantic Ocean near the United States coastline, operation **819** yielded at least 90-95% MCO_3 and at least 90-95% recovered CPI **810**.
(47) A first experimental example of CO_2/seawater mineralization was carried out according to process **801** using a CPI where R=cyclohexyl and the starred bond was to n-butyl (R′). At operation **813**, 150 mg CPI was suspended as a powder in CH_3CN, and air containing CO_2 was fed into the reaction mixture from a balloon inflated by exhaled air, which is a dilute source of CO_2 and water (˜4% CO_2 and ˜6% H_2O). The resulting formation of the soluble CPI-H^+/HCO_3^- salt occurred within about 10-20 minutes of

adding the exhaled air. Seawater was then added to the reaction mixture at operation **816**. The seawater was collected from Island Beach State Park, New Jersey.

(48) The addition of the seawater resulted in formation of CaCO.sub.3, MgCO.sub.3, and SrCO.sub.3 precipitate (13 mg, ˜95% yield MCO.sub.3) which was removed via filtration and drying of the aqueous phase. The organic phase solution of CPI-H.sup.+/X.sup.− salt was then concentrated by solvent evaporation. Toluene and approximately 1 M Na.sub.2CO.sub.3(aq) were added to the concentrated organic phase solution in order to recover the CPI at operation **909**. The recovery yielded 144 mg CPI (˜96% recovery).

(49) A second experimental example of CO.sub.2/seawater mineralization was carried out on a larger scale according to process **801** using a CPI where R=cyclohexyl and the starred bond was to n-butyl (R′). At operation **813**, 1.20 g CPI was suspended as a powder in CH.sub.3CN, and gaseous CO.sub.2 was fed into the reaction mixture. The resulting formation of the soluble CPI-H.sup.+/HCO.sub.3.sup.− salt was observed within about 10-20 minutes of adding the exhaled air. Seawater from Island Beach State Park, NJ, was then added to the reaction mixture (operation **816**).

(50) The addition of seawater resulted in formation of 105 mg MCO.sub.3 (CaCO.sub.3, MgCO.sub.3, and SrCO.sub.3) precipitate (˜90% yield), which was removed by filtration and drying of the aqueous phase. The organic phase solution of CPI-H.sup.+/X.sup.− salt was then concentrated by solvent evaporation. Toluene and approximately 1 M Na.sub.2CO.sub.3(aq) were then added to the organic phase solution in order to recover the CPI (operation **819**). This biphasic recovery yielded 1.16 g CPI (˜97% recovery).

(51) A third experimental example of CO.sub.2/seawater mineralization was carried out according to process **801** using the following polynorbornene CPI:

(52) ##STR00001##

where n is an integer greater than 1 and R=cyclohexyl. At operation **813**, 100 mg polynorbornene CPI was added to CH.sub.3CN, and about 101 Pa (1 atm) CO.sub.2 was fed into the reaction mixture. The reaction with CO.sub.2 resulted in formation of soluble polynorbornene with pendant CPI-H.sup.+/HCO.sub.3.sup.− groups. Seawater from Island Beach State Park, NJ, was then added to the reaction mixture.

(53) The addition of seawater (operation **816**) resulted in formation of MCO.sub.3 (CaCO.sub.3, MgCO.sub.3, and SrCO.sub.3) precipitate (11 mg, ˜95% yield) which was removed via filtration and drying of the aqueous phase. The organic phase was then concentrated and combined with toluene and approximately 1 M Na.sub.2CO.sub.3(aq) in order to recover the polynorbornene CPI (97% polynorbornene CPI recovery) at operation **819**.

(54) FIGS. **9**A and **9**B are chemical reaction diagrams illustrating generic processes **900** and **901** of CPI-facilitated polymerization, according to some embodiments. Processes **900** and **901** may be used to utilize and/or store captured CO.sub.2. For example, large scale processes may be used for DAC, CO.sub.2 storage, and/or conversion of CO.sub.2 into useful polymers. In FIGS. **9**A and **9**B, the reaction between CPI **810** and CO.sub.2 to form the adduct **820** is not shown. However, the adduct can be formed using substantially the same techniques as operation **803** or **813** (FIGS. **8**A and **8**B), e.g., by adding CO.sub.2 to a suspension of CPI **810** in acetonitrile to form the soluble adduct **820**. Like in FIGS. **8**A and **8**B, the CPI adduct **820** is illustrated with a starred bond, which can be a covalent bond, e.g., to a carbon atom of R′ (see, e.g., FIGS. **3-4**B), a metal complex, a polymer/oligomer repeat unit (see, e.g., FIGS. **5-7**), etc. In some embodiments, at least one R group may be replaced by a bond to a metal complex or polymer repeat unit as well.

(55) In process **900** (FIG. **9**A), the CPI-CO.sub.2 adduct **820** may act as a stoichiometric source of CO.sub.2 to form a polycarbonate or a polyurethane. In some embodiments, compound **903** is a diol or diamine. Herein, the abbreviation “Nu” on compound **903** represents a nucleophile, such as —O—, —N(H)—, or —N(R′)—, where R.sup.1=an organic substituent. That is, compound **903** can be a diol (when —NuH is a hydroxyl group (—OH)) or a diamine (when —NuH is a primary or secondary amine (respectively, —NH.sub.2 or —NHR.sup.1)). The portion of compound **903** represented by a circle labeled “A” in FIG. **9**A can be, e.g., a cyclic or acyclic organic group (see, e.g., FIGS. **11**A, **11**B, and **13**).

(56) In some embodiments, compound **906** is an aryl halide or aryl sulfonate. On compound **906**, X represents a leaving group, such as a halide or sulfonate (e.g., where X.sup.−=chloride, bromide, iodide, mesylate (OMs), tosylate (OTs), etc.). In various embodiments, the portion of compound **906** represented by a circle labeled “B” in FIG. **9**A can be any appropriate aromatic group (see e.g., FIG. **10**). When compound **903** is a diol, the product of process **900** can be a polycarbonate **909**. When compound **903** is a diamine, the product of process **900** can be a polyurethane **909**. Process **900** also results in formation of a CPI conjugate acid salt (CPI-H.sup.+/X.sup.−) **923** where X-corresponds to the leaving group **906**. The polymer **909** can then be obtained/purified, and CPI **810** can be recovered from CPI-H.sup.+/X.sup.− **923**. This is illustrated in FIG. **9**C. Process **900** may be repeated at least once using the recovered CPI.

(57) Process **901** (FIG. **9**B) is an example polymerization substantially similar to process **900**, except that three monomer species (generic compounds **913**, **916**, and **906**) are used to form a polyurethane or polycarbonate (polymer **919**). The portions of compounds **913** and **916** respectively represented by circles labeled “A.sup.1” and “A.sup.2” in FIG. **9**B can be independently selected from organic groups such as the cyclic or acyclic organic groups

discussed with respect to compound **906**. In further embodiments (not shown), there can be more than one or two monomer species with nucleophile groups and/or more than one monomer species with leaving groups in process **901**, as will be understood by persons of ordinary skill in the art. Process **901** also results in formation of CPI-H.sup.+/X.sup.− **923**. The polymer **919** can then be obtained/purified, and CPI **810** can be recovered from CPI-H.sup.+/X.sup.− **923**. This is illustrated in FIG. **9**C. Process **901** may be repeated at least once using the recovered CPI.

(58) FIG. **9**C is a chemical reaction diagram illustrating a process **902** of CPI recovery **900** after a CPI-facilitated polymerization (e.g., process **900** or **901**), according to some embodiments. FIG. **9**C illustrates recovery of CPI **810** from the conjugate acid salt **923** and purification of polymer **909** (FIG. **9**A) from the products of operation **900**. However, process **902** can also be used to obtain CPI **810** and polymer **919** after process **901** (FIG. **9**B) in various embodiments. Process **902** can be a biphasic recovery process, which may be similar to that of processes discussed in greater detail above with respect to FIGS. **2** and **8**B (basification of CPI-H.sup.+).

(59) In process **902**, a mixture (e.g., a CH.sub.3CN solution) of CPI/H.sup.+/X.sup.− **923** and polymer **909** can be combined with an aqueous alkaline solution (e.g., about 1 M Na.sub.2CO.sub.3(aq) or about 20-30 vol % NH.sub.4OH.sub.(aq)) and a non-polar solvent (e.g., toluene, hexanes, or a combination thereof). The resulting organic phase **914** can be separated from an aqueous phase (not shown) containing X.sup.− salt (e.g., NaX) and other water soluble reaction components (e.g., Na.sub.2CO.sub.3, and NaHCO.sub.3). The organic phase **914** can include CPI **810** (soluble) and polymer **909** (insoluble). The polymer **909** precipitate can be obtained (e.g., filtered, purified, etc.) from the organic phase **914** using techniques known to those of ordinary skill in the art (not shown). The CPI **810** can be recovered from the remaining organic solution and optionally reused for additional CO.sub.2 capture (e.g., process **200** illustrated in FIG. **2**). In some embodiments, CPI recovery processes such as those discussed above can be carried out after one or more CO.sub.2 capture/activation cycles, thereby allowing reuse of CPI over multiple polymerization cycles.

(60) FIG. **10** is a chemical reaction diagram illustrating example processes **1010** and **1020** of CPI-facilitated polymerization, according to some embodiments. Processes **1010** and **1020** may be carried out according to generic process **900** in some embodiments. In process **1010**, the CPI-CO.sub.2 adduct **820** reacts with 1,8-diamino-3,6-dioxaoctane (amino-PEG2-amine) and 1,4-bis(bromomethyl)benzene in acetonitrile to form polyurethane (where n is an integer greater than 1). In process **1020**, the CPI-CO.sub.2 adduct reacts with ˜400 Da polyethylene glycol and 1,4-bis(bromomethyl)benzene in acetonitrile to form polycarbonate (where n is an integer greater than 1). As discussed above, CPI **810** (not shown in FIG. **10**) can be recovered and reused after a first polymerization cycle **1010** or **1020**.

(61) FIGS. **11**A and **11**B are tables **1100** and **1101** illustrating chemical structures corresponding to experimental examples of CPI-facilitated polymerizations. Tables **1100** and **1101** show examples of monomers (compounds **903** and **906**) used in experimental examples carried out using techniques substantially similar to those of process **900**, as well as the resulting polymers 1A-1H (polymer **909**) formed in the examples.

(62) FIG. **12** is a table **1200** illustrating experimental data characterizing the polymers 1A-1H illustrated in FIGS. **11**A and **11**B. The values shown in table **1200** (mean molecular weight, dispersity, decomposition temperature, and glass transition temperature) were experimentally obtained for each polymer 1A-1H using known techniques.

(63) FIG. **13** is a table **1300** illustrating chemical structures corresponding to an experimental example of CPI-facilitated polymerization, according to some embodiments. Table **1300** shows monomers (corresponding to compounds **913**, **916**, and **906**) reacted with activated CO.sub.2 in an experimental example carried out using techniques substantially similar to those of process **901**, as well as the resulting polymer 2A (polymer **919**) formed in the reaction.

(64) FIG. **14** is a table illustrating experimental data characterizing a polymer product formed in the experimental example of FIG. **13**, according to some embodiments. The values shown in table **1400** (mean molecular weight, dispersity, decomposition temperature, and glass transition temperature) were experimentally obtained for polymer 2A using techniques known to those of ordinary skill in the art.

(65) A wide variety of CPI-containing materials ("CPI materials") can be formed based on CPI small molecules (see, e.g., FIGS. **3**-**4**B), polymers with CPI pendant groups (see, e.g., FIGS. **5**-**7**), other polymers and polymer networks (see below), etc. These CPI materials can include CPI moieties having the following structure:

(66) ##STR00002##

where each starred bond can be a covalent bond to, e.g., a carbon atom of an organic moiety (R) or a polymer repeat unit, as discussed in greater detail above. The CPI moieties can allow the materials to be used for CO.sub.2 capture/activation technologies on a variety of scales. These CPI materials may be recovered after one or more CO.sub.2 capture/activation processes using, for example, a biphasic recovery process (e.g., operation **819** of process **801**, illustrated in FIG. **8**B).

(67) In some embodiments, the polymers can be multi-functional polymers for capturing CO.sub.2 and transforming the CO.sub.2 into new chemicals (e.g., metal carbonates). Copolymerization of the building blocks (e.g., monomers or oligomers) with other functional monomers can be used to tune both CO.sub.2 uptake and processability of the

final polymers. Various macromolecular architectural considerations may also be used for tuning these properties. Examples of polymer architectures may include linear, branched, dendritic, bottle brush, surface-grafted, etc. Techniques for automated polymerization, high-throughput characterization, predictive modeling, etc. may be employed to facilitate selection of material compositions. Through selection of monomers/oligomers used in these processes, both homogeneous and segmented morphologies can be generated, allowing control over air permeation, modulus, hydrophilic/hydrophobic balance, and other key structural features.
(68) In some embodiments, CPI polymers can be processed in modular architectures such as particles, suspensions, membranes, fibers, gels, coatings, etc. These CPI materials can be formed by polymerization of CPIs and/or TAC ion derivatives (see above). CPI and/or TAC molecules may also be used to functionalize polymeric materials such as particles, suspensions, membranes, fibers or other solid materials, gels, coatings, etc. in some embodiments. For example, nano- or microparticles can be formed from polymers with CPI/TAC pendant groups and/or surface-functionalized with CPI molecules. CPI surfactants may be used as coatings as well. In some embodiments, a CPI-fiber material can be formed by polymerizing or copolymerizing CPI monomers, functionalizing a polymer fiber with CPIs, crosslinking a polymer fiber with CPIs, coating a fibrous material (e.g., woven or knit fabrics, paper or cloth filters, filler/insulation materials, etc.) with a CPI small-molecule, oligomer, or polymer coating, or any other techniques for incorporating molecules such as CPIs into fibrous materials.
(69) The CPI materials can be used in DAC technologies to sequester and upcycle (e.g., activate or release in response to a stimulus) CO_2. For example, the CPI materials may be exposed to gaseous CO_2 in a solvent or solvent-free environment. In examples involving CPI-CO_2 adducts formed in a solvent-free environment, activation of the CO_2 and recovery of the CPI may be carried out in appropriate solvents, and the recovered CPI can optionally be dried and reused for CO_2 capture in the solvent-free environment. The CPI-facilitated CO_2 capture and activation can allow the CPI materials to facilitate mineralization or polymerization reactions such as those discussed above. The reaction products (e.g., aqueous phase metal carbonates and organic phase CPI-H^+/Cl^- conjugate acid) can then be separated, and the CPI materials can be recovered for additional CO_2 capture/activation. For example, a biphasic process with the conjugate acid material (see, e.g., FIG. **8**B) can be used to recover the CPI material.
(70) The formation of CPI-CO_2 adducts and subsequent recovery of CPI involves transformation from a polar zwitterionic form (adduct) to a non-polar neutral free-base form (CPI). This chemical change may be used to drive macromolecular phase transformations reversibly induced by CO_2 capture followed by activation or release. For example, phase transitions of ABA triblock copolymers having hydrophobic A-blocks derived from CPI and a center B-block derived from hydrophilic polymers may be used. These triblock copolymers can be prepared from difunctional hydrophilic blocks. In some embodiments, the R-groups on CPI can be used to drive phase changes. For example, morpholine-substituted CPIs are likely to be hydrophilic, and may therefore be used as the mid-block in order to increase the number of CO_2 capture sites within the entire triblock.
(71) For example, a sol-gel CO_2 capture/release process (not shown) can be carried out using the aforementioned triblock copolymer with CPI pendant groups. The frustrated outer hydrophobic blocks fold back to form a hydrophobic pocket or “flower-like” micelle. Above ˜3-5% concentration in an aqueous solution, mild agitation can cause the hydrophobic chains to interdigitate, thereby connecting the micelles and producing a hydrogel. CO_2 capture by passing air through the gel can cause a phase-change from a gel to a sol. This is because the CPI-CO_2 adduct is hydrophilic and its formation causes unfolding of the micelles. The sol copolymer can be captured and converted back to the gel upon release of the CO_2.
(72) Various types of apparatus may be used in mediating absorption for DAC. For example, polymeric CPI materials for CO_2 capture/transfer (e.g., for mineralization or polymerization) may be employed in a packed bead reactor, trayed adsorption column, spray tower, spray dryer, etc. (see below). Techniques for gas-liquid mass transfer known to those of ordinary skill may be employed, and parameters such as flow rates, temperatures, concentrations, residence times, packing or tray types, nozzle design, droplet size (in spray methods) can be tuned.
(73) In a packed bead reactor, there can be an absorption column that uses polymeric micro- and/or nanoparticles as a CPI-functionalized solid support resin. The absorption column can be packed with CPI-functionalized particles, and a CO_2-containing gas phase (e.g., atmospheric gas) can be passed through the column until CO_2 breakthrough is observed. Following the CO_2 exposure, the column can be detached, regenerated, and the gas released using techniques such as those discussed above (e.g., by photoredox/irradiation, mechanical force such as sonication, etc.). In some embodiments, CPI-functionalized particles can be formed using CPI-styrene monomers behaving as surfactants. In these instances, polymerization with a core derived from a hydrophobic styrene and various concentrations of divinylbenzene (DVB), can generate highly crosslinked particles by mini-emulsion polymerization.
(74) In a trayed absorption column, a CO_2-containing gas can be continually introduced at the bottom of the column while a CO_2-absorbing liquid, which includes a CPI-functionalized small-molecule or polymer, is introduced at the top of the column. As the gas and liquid phases mix in the column, the gas can percolate on trays positioned in the column to allow sufficient residence time for gas absorption into the liquid phase. The scrubbed gas

can then be collected at the top of the column, and the CPI-CO_2-containing liquid can be collected at the bottom of the column for further downstream processing (e.g., including upcycling/mineralization and/or release of the captured CO_2).
(75) A spray tower can utilize an aqueous solution or emulsion of a CPI-functionalized polymer or CPI small molecules. This CPI-containing liquid can be sprayed from the top of the tower into a CO_2-containing gas. As in the trayed absorption column, the solution containing the captured CO_2 (CPI-CO_2 adducts) can then be collected at the bottom for further downstream processing.
(76) A spray dryer can be similar to the spray tower. For example, a controlled mist of the CPI-containing liquid can be introduced into a tower or column concurrently with CO_2-containing air (e.g., heated CO_2-containing air). In this configuration, the liquid can be heated to ensure complete evaporation of the liquid phase and produce a solid aerosol and “wet” air. A cyclone separator can be used to disengage the solid material (the small-molecule or polymeric CPI-CO_2 adduct) from the flowing air.
(77) Various embodiments of the present disclosure are described herein with reference to the related drawings, where like numbers refer to the same component. Alternative embodiments can be devised without departing from the scope of the present disclosure. It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present disclosure is not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. As an example of an indirect positional relationship, references in the present description to forming layer “A” over layer “B” include situations in which one or more intermediate layers (e.g., layer “C”) is between layer “A” and layer “B” as long as the relevant characteristics and functionalities of layer “A” and layer “B” are not substantially changed by the intermediate layer(s).
(78) The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus. Further, the word “providing” as used herein can refer to various actions such as creating, purchasing, obtaining, synthesizing, making available, etc. or combinations thereof.
(79) As used herein, the articles “a” and “an” preceding an element or component are intended to be nonrestrictive regarding the number of instances (i.e., occurrences) of the element or component. Therefore, “a” or “an” should be read to include one or at least one, and the singular word form of the element or component also includes the plural unless the number is obviously meant to be singular.
(80) As used herein, the terms “invention” or “present invention” are non-limiting terms and not intended to refer to any single aspect of the particular invention but encompass all possible aspects as described in the specification and the claims.
(81) Unless otherwise noted, ranges (e.g., time, concentration, temperature, etc.) indicated herein include both endpoints and all numbers between the endpoints. Unless specified otherwise, the use of a tilde (˜) or terms such as “about,” “substantially,” “approximately,” “slightly less than,” and variations thereof are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of ±8% or 5%, or 2% of a given value, range of values, or endpoints of one or more ranges of values. Unless otherwise indicated, the use of terms such as these in connection with a range applies to both ends of the range (e.g., “approximately 1 g-5 g” should be interpreted as “approximately 1 g-approximately 5 g”) and, in connection with a list of ranges, applies to each range in the list (e.g., “about 1 g-5 g, 5 g-10 g, etc.” should be interpreted as “about 1 g-about 5 g, about 5 g-about 10 g, etc.”).
(82) As discussed above, CPIs and other compounds herein include R groups (e.g., R, R′, and R^x, where x is an integer), which can be any appropriate organic substituent known to persons of ordinary skill. In some embodiments, the R groups can include substituted or unsubstituted aliphatic groups. As used herein, the term “aliphatic” encompasses the terms alkyl, alkenyl, and alkynyl.
(83) As used herein, an “alkyl” group refers to a saturated aliphatic hydrocarbon group containing from 1 to 20 (e.g., 2 to 18, 2 to 8, 2 to 6, or 2 to 4) carbon atoms. An alkyl group can be straight, branched, cyclic, or any combination thereof. Examples of alkyl groups include, but are not limited to, methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-heptyl, or 2-ethylhexyl. An alkyl group can be substituted with one or more substituents or can be multicyclic as set forth below. Unless specified otherwise, the term “alkyl,” as well as derivative terms such as “alkoxy” and “thioalkyl,” as used herein, include within their scope, straight chain, branched chain, and cyclic moieties.
(84) As used herein, an “alkenyl” group refers to an aliphatic carbon group that contains from 2 to 20 (e.g., 2 to 18, 2

to 8, 2 to 6, or 2 to 4) carbon atoms and at least one double bond. Like an alkyl group, an alkenyl group can be straight, branched, or cyclic, or any combination thereof. Examples of an alkenyl group include, but are not limited to, allyl, isopropenyl, 2-butenyl, and 2-hexenyl. An alkenyl group can be substituted with one or more substituents as set forth below.

(85) As used herein, an “alkynyl” group refers to an aliphatic carbon group that contains from 2 to 20 (e.g., 2 to 18, 2 to 8, 2 to 6, or 2 to 4) carbon atoms and has at least one triple bond. Like an alkyl group, an alkynyl group can be straight, branched, or cyclic, or any combination thereof. Examples of an alkynyl group include, but are not limited to, propargyl and butynyl. An alkynyl group can be substituted with one or more substituents as set forth below.

(86) The term “alkylthio” includes straight-chain alkylthio, branched-chain alkylthio, cycloalkylthio, cyclic alkylthio, heteroatom-unsubstituted alkylthio, heteroatom-substituted alkylthio, heteroatom-unsubstituted Cn-alkylthio, and heteroatom-substituted Cn-alkylthio. In some embodiments, lower alkylthios are contemplated.

(87) The term “haloalkyl” refers to alkyl groups substituted with from one up to the maximum possible number of halogen atoms. The terms “haloalkoxy” and “halothioalkyl” refer to alkoxy and thioalkyl groups substituted with from one up to five halogen atoms.

(88) As described herein, compounds of the present disclosure can optionally be substituted with one or more substituents, such as are illustrated generally above, or as exemplified by particular classes, subclasses, and species of the present disclosure. Each substituent of a specific group may further be substituted with one to three of, for example, halogen, cyano, sulfonyl, sulfinyl, carbonyl, oxoalkoxy, hydroxy, amino, nitro, aryl, haloalkyl, and alkyl. For instance, an alkyl group can be substituted with alkyl sulfonyl and the alkyl sulfonyl can be optionally substituted with one to three of halogen, cyano, sulfonyl, sulfinyl, carbonyl, oxoalkoxy, hydroxy, amino, nitro, aryl, haloalkyl, and alkyl.

(89) In general, the term “substituted” refers to the replacement of hydrogen radicals in a given structure with the radical of a specified substituent. Specific substituents are described above in the definitions and below in the description of compounds and examples thereof. Unless otherwise indicated, an optionally substituted group can have a substituent at each substitutable position of the group, and when more than one position in any given structure can be substituted with more than one substituent selected from a specified group, the substituent can be either the same or different at every position. A ring substituent, such as a hetero cycloalkyl, can be bound to another ring, such as a cycloalkyl, to form a spiro-bicyclic ring system, e.g., both rings share one common atom. As one of ordinary skill in the art will recognize, combinations of substituents envisioned by this present disclosure are those combinations that result in the formation of stable or chemically feasible compounds.

(90) Modifications or derivatives of the disclosed compounds are contemplated as being useful with the methods and compositions of the present disclosure. Derivatives may be prepared and the properties of such derivatives may be assayed for their desired properties by any method known to those of skill in the art. In certain aspects, “derivative” refers to a chemically modified compound that still retains the desired effects of the compound prior to the chemical modification.

(91) The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments described. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

Claims

1. A process of generating polymers, comprising: reacting carbon dioxide (CO.sub.2) with a cyclopropenimine (CPI); and reacting monomers with a product of the reacting, wherein the monomers comprise a monomer having nucleophilic functional groups.
2. The process of claim 1, wherein the monomers further comprise a monomer having halide functional groups or sulfonate functional groups.
3. The process of claim 1, wherein the nucleophilic functional groups comprise hydroxyl groups.
4. The process of claim 3, wherein the reacting the monomers with the product generates a polycarbonate.
5. The process of claim 1, wherein the nucleophilic functional groups comprise amine groups.
6. The process of claim 5, wherein the reacting the monomers with the product generates a polyurethane.
7. The process of claim 1, wherein the reacting the CO.sub.2 with the CPI comprises adding the CO.sub.2 to a suspension of the CPI in a polar organic solvent.
8. The process of claim 1, wherein the reacting the CO.sub.2 with the CPI comprises obtaining the CO.sub.2 from air.
9. The process of claim 1, wherein the CPI is incorporated into a support material.

10. The process of claim 1, wherein the CPI is a superbase.
11. The process of claim 1, wherein the CPI comprises electron-donating amine substituents and wherein the product of the reacting the CO.sub.2 with the CPI is a CPI-CO.sub.2 adduct comprising a tris(amino)cyclopropenium (TAC) ion that is stabilized by the electron-donating amine substituents.
12. The process of claim 11, wherein the reacting the product with the monomers results in formation of a polymer and a conjugate acid of the CPI-CO.sub.2 adduct, wherein the polymer is polycarbonate or polyurethane.
13. The process of claim 12, further comprising recovering the CPI from the conjugate acid in a biphasic process comprising mixing the conjugate acid with an aqueous alkaline solution and a non-polar organic solvent.
14. The process of claim 11, wherein the CPI-CO.sub.2 adduct has the following structure: ##STR00003## wherein each R is an organic substituent, and wherein the starred bond is to a carbon atom.
15. The process of claim 1, wherein the CPI has the following structure: ##STR00004## wherein each R is an organic substituent, and wherein the starred bond is to a carbon atom.
16. The process of claim 15, wherein the starred bond is a bond to a polymer repeat unit.
17. The process of claim 15, wherein the organic substituent is cyclohexyl.
18. The process of claim 1, wherein the CPI is a functional group bound to a polystyrene solid support.
19. The process of claim 1, wherein the reacting the CO.sub.2 with the CPI comprises: providing an absorption column packed with CPI-functionalized particles; and passing atmospheric gas containing the CO.sub.2 through the absorption column.